

ASSIGNMENT

1. Difference Between Static Testing and Dynamic Testing

In software development, testing can be broadly divided into static and dynamic testing.

Static Testing

Static testing is done without running the program. In this type of testing, we check documents like requirement specifications, design documents, or even the source code to identify mistakes at an early stage. The main aim is to prevent defects before execution.

Examples:.. Code Review – Developers examine each other's code to find logical or syntax errors.

. Walkthrough – The author explains the document or code step-by-step to the team for feedback.

Dynamic Testing

Dynamic Testing is performed by actually executing the software. It helps us verify whether the system works as expected.

Examples:

Unit Testing – Testing individual modules separately.

System Testing – Testing the complete system as a whole.

2. Principles of Software Testing and Importance of Early Testing

Software testing follows some important principles:

1. Testing can show the presence of defects, but not prove that the software is completely error-free.

2. It is impossible to test everything, so testing should be prioritized.
3. Testing should start as early as possible.
4. Most defects are usually found in a small number of modules.
5. Testing methods depend on the type of application.
6. A system without errors may still fail if it does not meet user requirements.

Why Early Testing is Cost-Effective

Early testing is considered cost-effective because fixing a defect in the requirement or design phase is much easier and cheaper than fixing it after development or deployment. If an error is found late, it may require redesigning, recoding, and retesting the entire system. Therefore, early testing saves time, money, and effort while improving software quality.

3. Waterfall Model and V-Model in SDLC

Waterfall Model

The Waterfall Model is a traditional and sequential development model. Each phase must be completed before moving to the next one.

Phases include:

- Requirement Analysis
- Design
- Implementation
- Testing
- Deployment
- Maintenance

In this model, testing happens only after the development phase is completed. Because of this, defects may be identified late.

V-Model

The V-Model is an improved version of the Waterfall model. It connects each development phase with a corresponding testing phase.

For example:

- Requirements → Acceptance Testing
- System Design → System Testing
- Architecture Design → Integration Testing
- Module Design → Unit Testing

The V-Model gives equal importance to verification and validation, ensuring that testing is planned early in the project.

Difference

The main difference is that in the Waterfall model, testing is done after development, whereas in the V-Model, testing activities are planned alongside development. Therefore, the V-Model helps in early detection of defects.

4. Roles and Ceremonies in Agile Scrum Framework

Scrum is a popular Agile framework used for managing software projects.

Key Roles:

- Product Owner – Responsible for defining requirements and managing the product backlog.
- Scrum Master – Ensures that the team follows Scrum practices and removes obstacles.
- Development Team – Responsible for developing and delivering the product increment.

Scrum Ceremonies:

- Sprint Planning – Planning the work for the sprint.
- Daily Scrum – Short daily meeting to discuss progress.
- Sprint Review – Demonstration of completed work.
- Sprint Retrospective – Discussion about improvements.

Importance of a Sprint

A Sprint is a fixed time period (usually 2–4 weeks) during which a usable product increment is created. It helps in delivering work faster, getting continuous feedback, and adapting to changes easily. Sprints make the development process flexible and efficient.

5. Requirement Traceability Matrix (RTM)

A Requirement Traceability Matrix (RTM) is a document used to map requirements with their corresponding test cases. It ensures that all requirements are properly tested and nothing is missed

Requirement ID	Requirement Description	Test Case ID	Test Case Description
R1	User must enter a valid email address	TC01	Verify email format validation
R2	Password must contain minimum 8 characters	TC02	Check password length validation

R3	All mandatory fields must be filled	TC03	Verify error message for empty fields
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